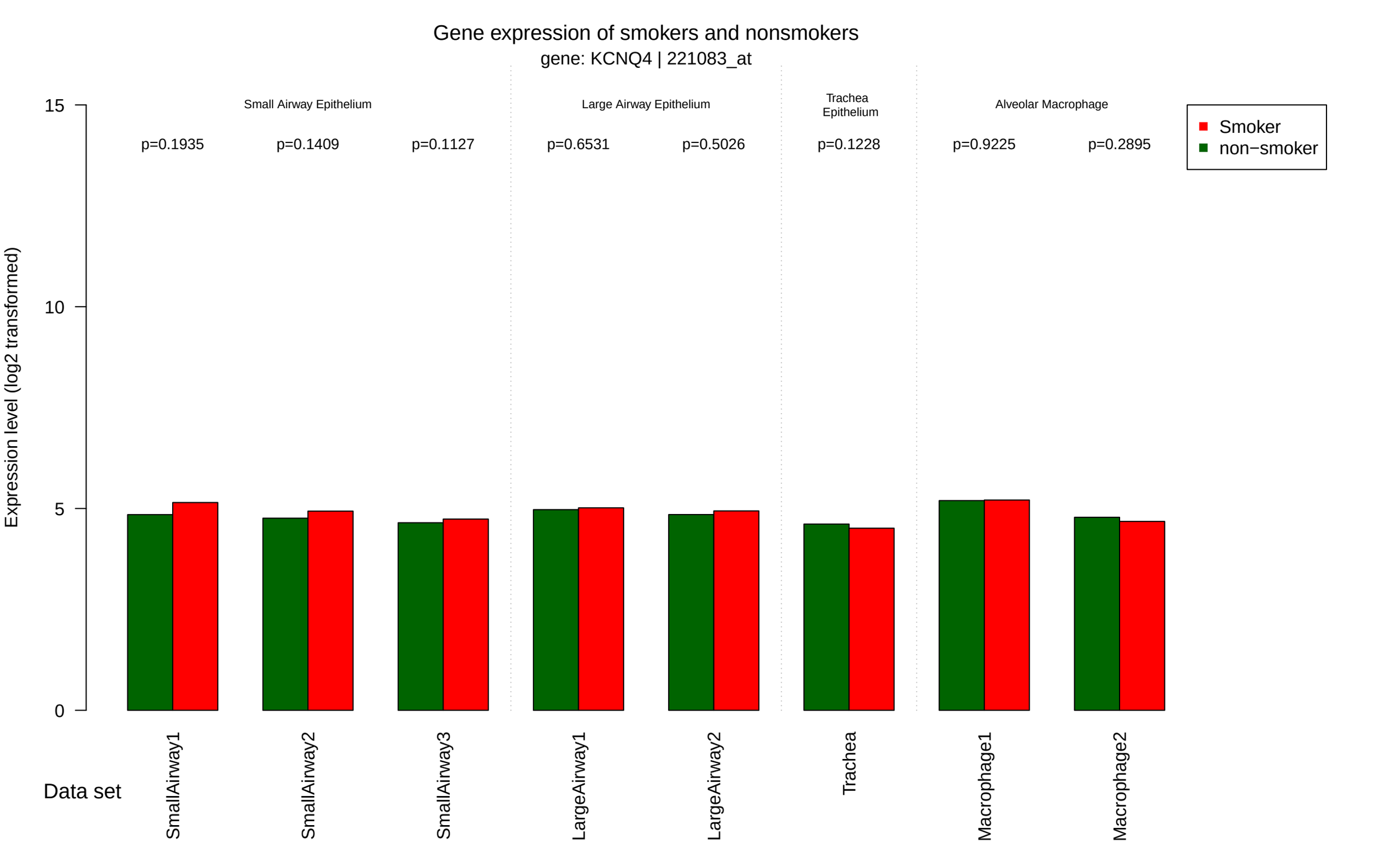
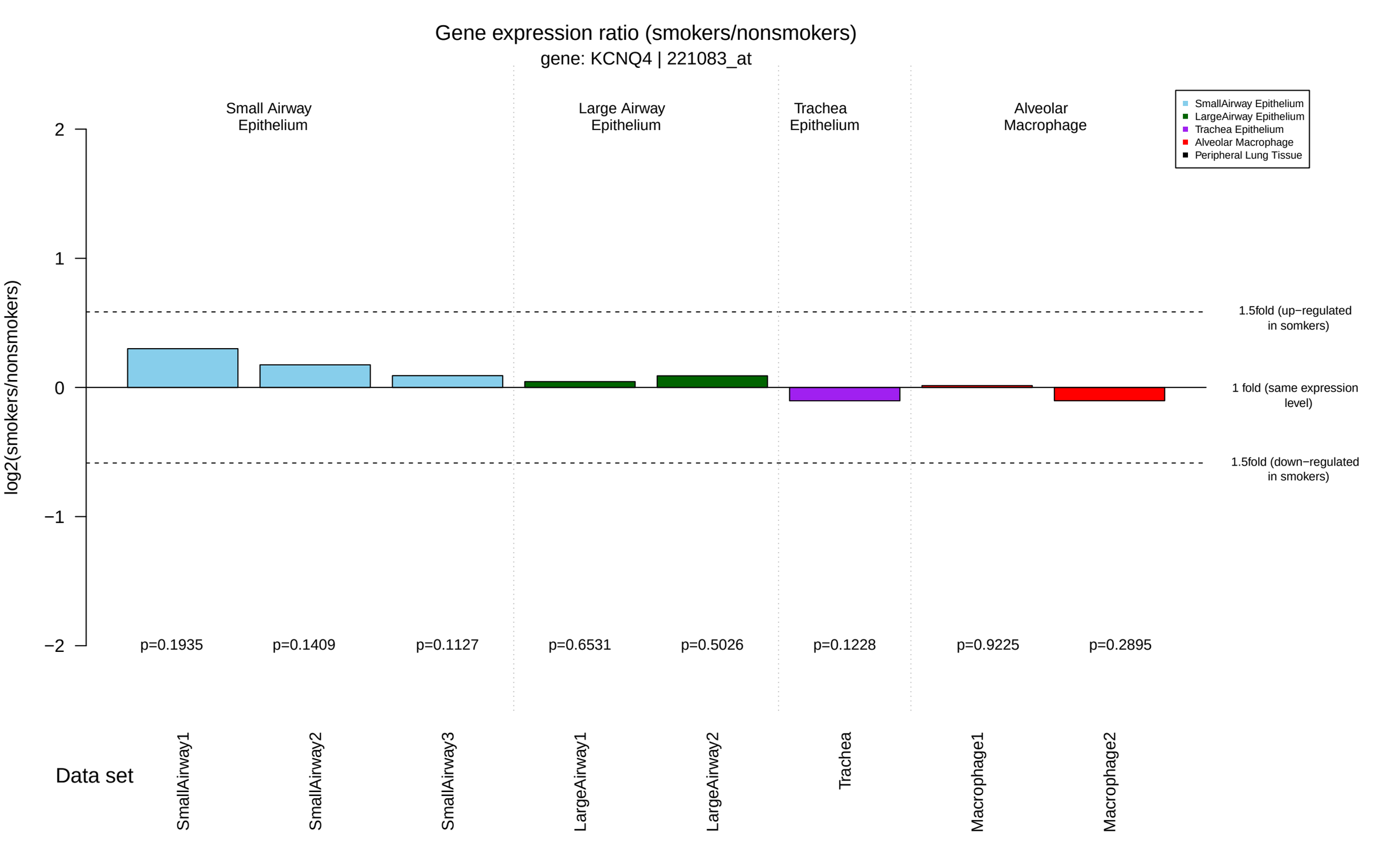
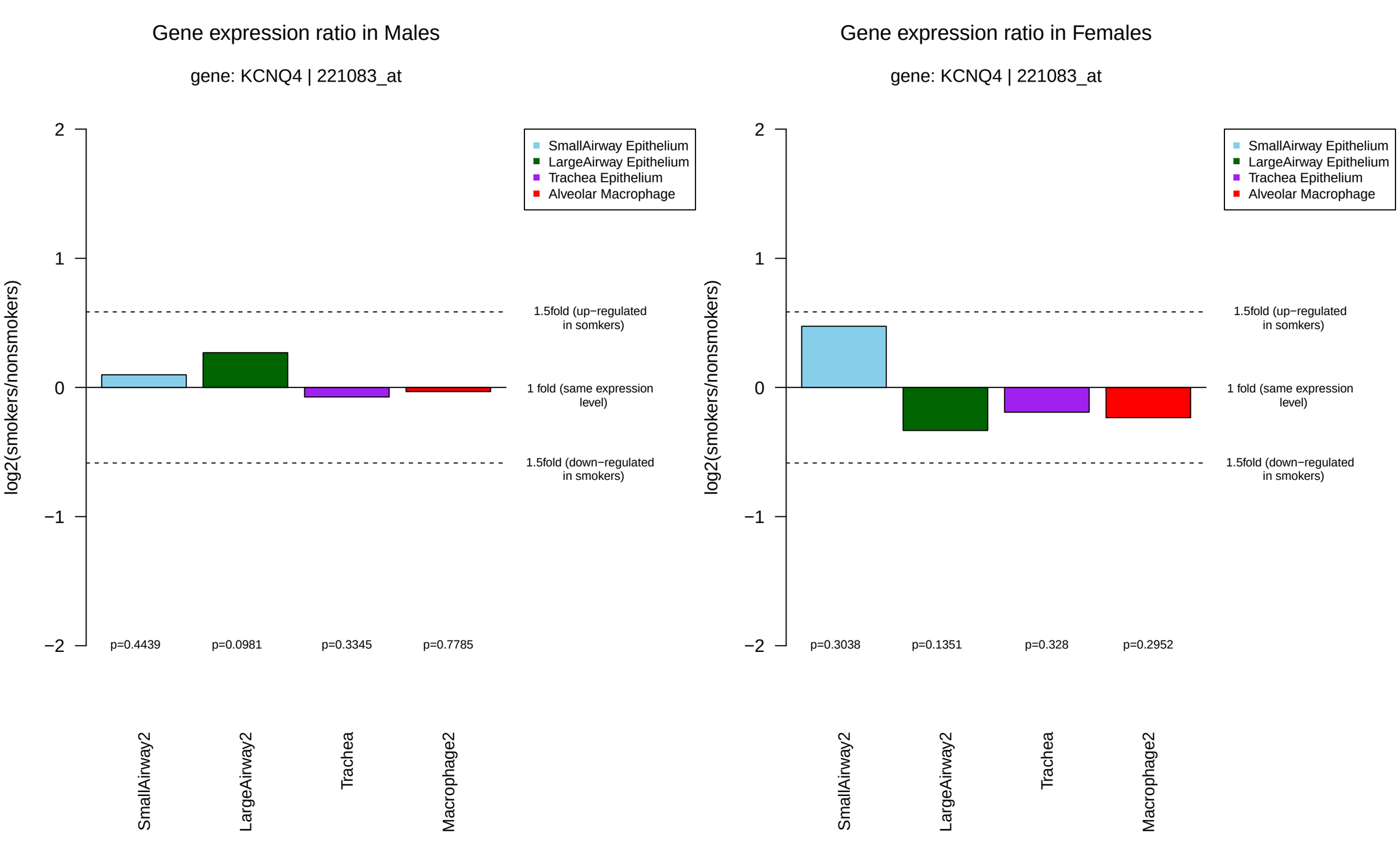
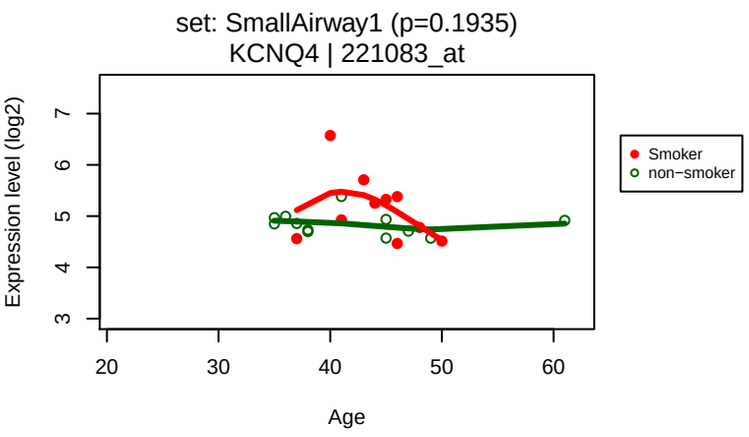
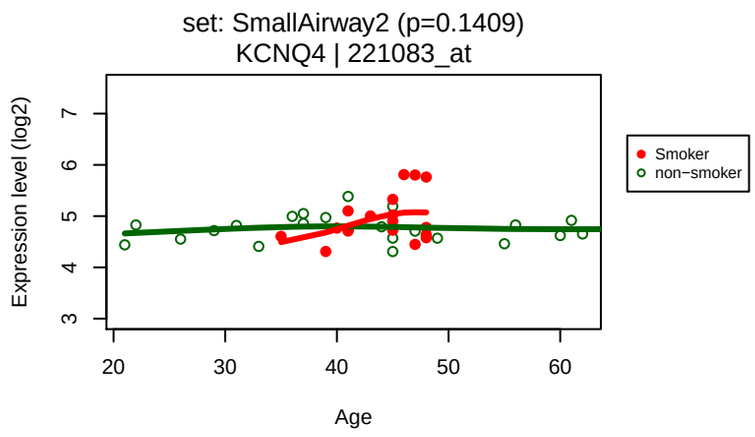
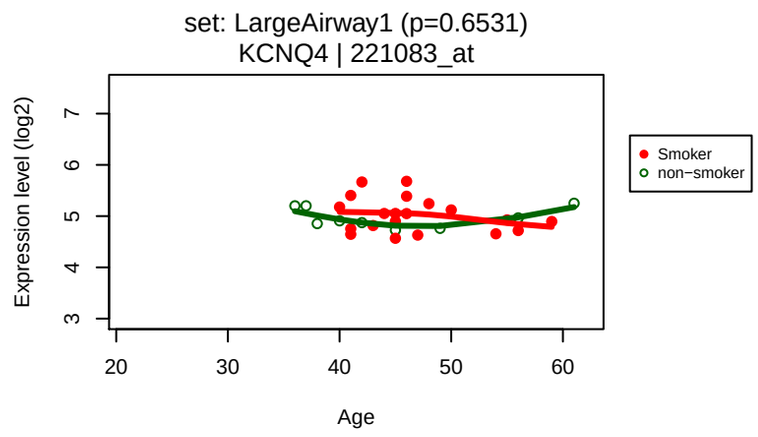

Gene expression of smokers and nonsmokers
gene: KCNQ4 | 221083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1935
p=0.1409
p=0.1127
p=0.6531
p=0.5026
p=0.1228
p=0.9225
p=0.2895
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

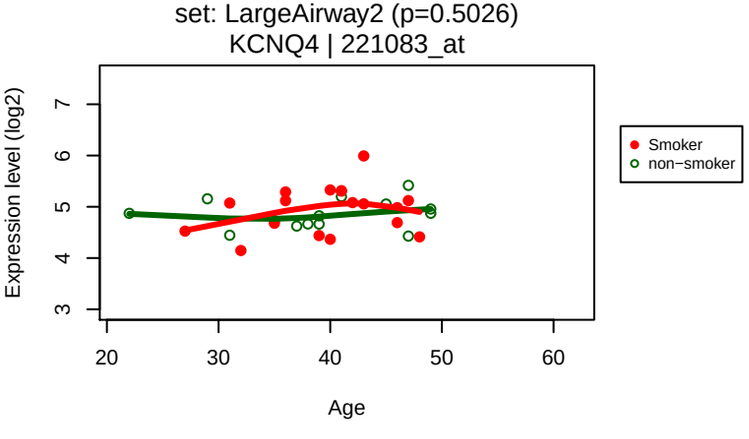

Gene expression ratio (smokers/nonsmokers)
gene: KCNQ4 | 221083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1935
p=0.1409
p=0.1127
p=0.6531
p=0.5026
p=0.1228
p=0.9225
p=0.2895
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

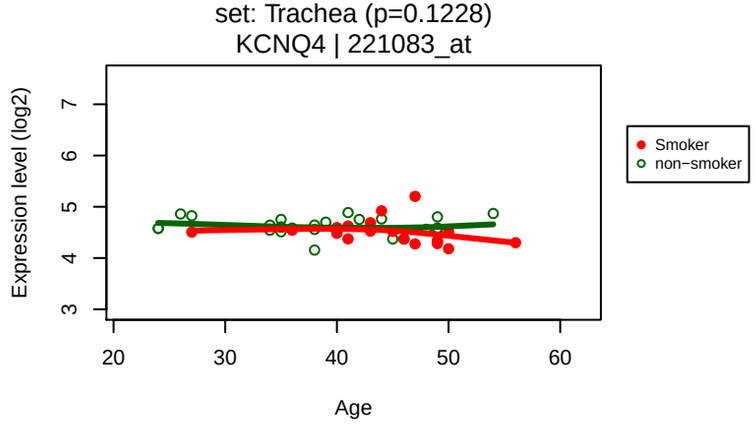

Gene expression ratio in Males
gene: KCNQ4 | 221083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4439
p=0.0981
p=0.3345
p=0.7785
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNQ4 | 221083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3038
p=0.1351
p=0.328
p=0.2952
SmallAirway2
LargeAirway2
Trachea
Macrophage2

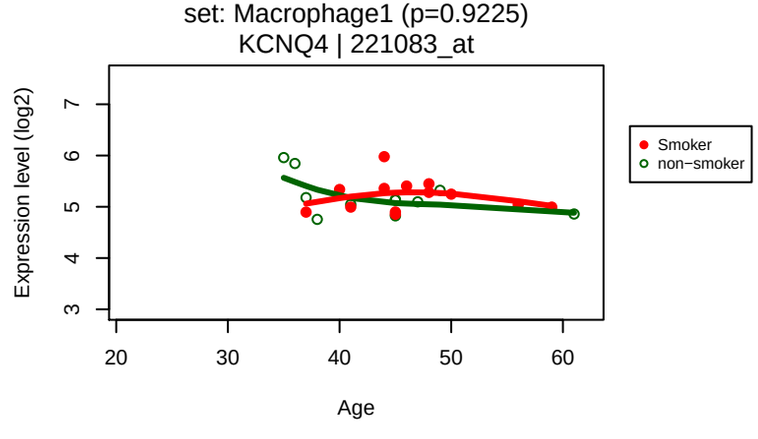

set: SmallAirway1 (p=0.1935)
KCNQ4 | 221083_at
Expression level (log2)
Age
Smoker
non−smoker

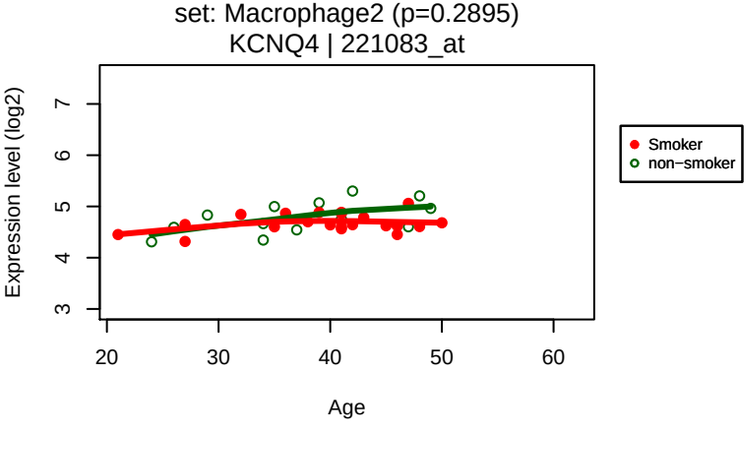

set: SmallAirway2 (p=0.1409)
KCNQ4 | 221083_at
Expression level (log2)
Age
Smoker
non−smoker

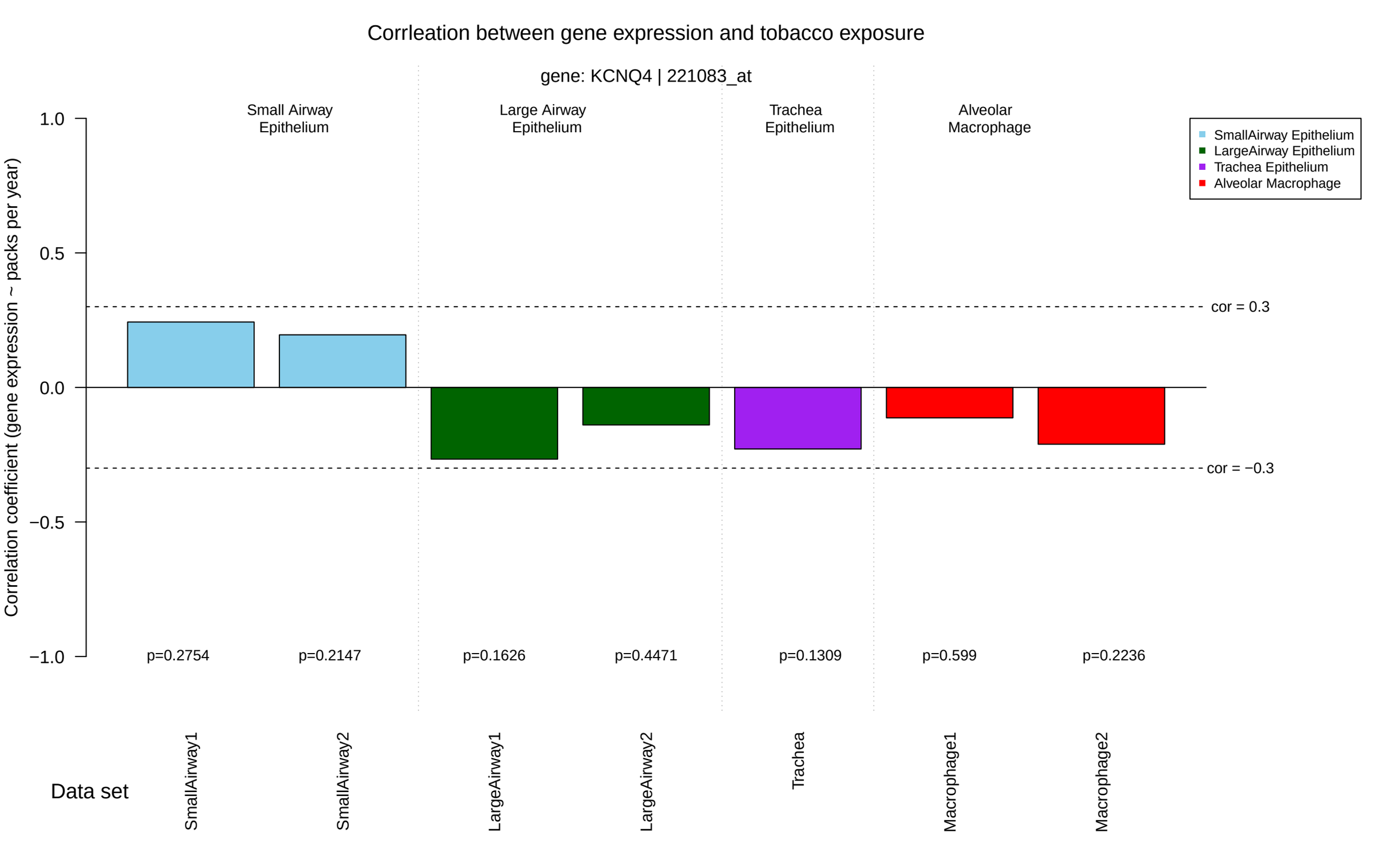

set: LargeAirway1 (p=0.6531)
KCNQ4 | 221083_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5026)
KCNQ4 | 221083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1228)
KCNQ4 | 221083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9225)
KCNQ4 | 221083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2895)
KCNQ4 | 221083_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNQ4 | 221083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2754
p=0.2147
p=0.1626
p=0.4471
p=0.1309
p=0.599
p=0.2236
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2